Project Presentation

In this project, my goal is to predict whether a bank customer will leave the bank or stay. This is called **Customer Churn Prediction**, and I use an **Artificial Neural Network (ANN)**.

First, I load and analyze the Churn Modelling dataset. I remove unnecessary columns such as RowNumber, CustomerId, and Surname.

Then, I convert categorical features into numerical values. I use **Label Encoding** for Gender and **One-Hot Encoding** for Geography. After preprocessing, I separate the features X from the target variable y, which is Exited.

Next, I split the data into training and testing sets and use **StandardScaler** to scale the features. I also save the encoders and scaler with Pickle so I can use the same preprocessing when making predictions later.

After that, I build the ANN using Keras. The network has two hidden layers with **64 and 32 neurons**, both using **ReLU**, and an output layer with **one neuron using Sigmoid** because this is a binary classification problem.

For training, I use the **Adam optimizer** and **Binary Focal Crossentropy** as the loss function. I also use **Early Stopping** to prevent unnecessary training and **TensorBoard** to monitor the training process.

Finally, I train the model, evaluate its performance, and save the trained model as model.h5.

In summary, the workflow is:

Load data → Preprocess data → Encode features → Split and scale data → Build ANN → Train the model → Monitor performance → Save the model.

The final goal is to predict whether a customer is likely to **leave the bank or stay**.

Next I put them into site with streamlit to make it easier for who wants to enter data.